

Tekstmodel

	model	initial values
1	$I = U_b / (R_3 + R_{1,2}/2)$	$t = 0 \text{ s}$
2	$U_{1,2} = I \cdot R_{1,2}/2$	$dt = 1 \times 10^{-4} \text{ s}$
3	$U_3 = I \cdot R_3$	$U_b = 18 \text{ V}$
4	$P_{1,2} = I^2 \cdot R_{1,2}/4$	$R_0 = 5.0 \Omega$
5	$P_3 = I^2 \cdot R_3$	$T_0 = 293 \text{ K}$
6	$T_{1,2} = T_{1,2} + (P_{1,2} - \varepsilon \sigma A \cdot T_{1,2}^4) \cdot dt / mc$	$\alpha = 5.2 \times 10^{-3} \text{ K}^{-1}$
7	$T_3 = T_3 + (P_3 - \varepsilon \sigma A \cdot T_3^4) \cdot dt / mc$	$\varepsilon \sigma A = 7.14 \times 10^{-14}$
8	$R_{1,2} = R_0 \cdot (1 + \alpha \cdot (T_{1,2} - T_0))$	$mc = 3 \times 10^{-5} \text{ J K}^{-1}$
9	$R_3 = R_0 \cdot (1 + \alpha \cdot (T_3 - T_0))$	$I = 0 \text{ A}$
10	$t = t + dt$	$R_{1,2} = 5.0 \Omega$
11	IF $t \geq 0.3$ THEN STOP ENDIF	$R_3 = 5.0 \Omega$ $U_{1,2} = 0 \text{ V}$ $U_3 = 0 \text{ V}$ $P_{1,2} = 0 \text{ W}$ $P_3 = 0 \text{ W}$ $T_{1,2} = 3 \times 10^2 \text{ K}$ $T_3 = 3 \times 10^2 \text{ K}$

Grafisch model

